Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/22/2017
PI Name:	Wotring, Virginia Ph.D.		
Project Title:	Inflight Pharmacokinetic and Pharmacodynamic Responses to Medications Commonly Used in Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:	PI formerly with Universities Space Research Association until fall 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	03/17/2016	**End Date:**	10/01/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: NASA deferred project to a later, unspecified time per PI and NSSC information. End date changed to 9/30/2017; using 10/01/2017 for reporting purposes (Ed., 12/29/17)		
Key Personnel Changes/Previous PI:	December 2017 report: NASA deferred project to a later, unspecified time. Activities ended. January 2017 report: Hired Kyla Cook, MS, as project manager/data analyst.		
COI Name (Institution):	Barger, Laura Ph.D. (Brigham And Women's Hospital) Derendorf, Hartmut Ph.D. (University of Florida) Basner, Mathias M.D., Ph.D. (University of Pennsylvania)		
Grant/Contract No.:	NNX16AG97G		
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Task Description:	NOTE (July 2016): Continuation of "Inflight Pharmacokinetic and Pharmacodynamic Responses to Medications Commonly Used in Spaceflight--Internal Project" with the same investigator, Dr. Virginia Wotring, due to PI move to Baylor College of Medicine (BCM). Crewmembers on spaceflight missions use medications to treat spaceflight-associated conditions like space adaptation syndrome as well as common ordinary complaints. They do so under the remote guidance of their flight surgeons, who base prescriptions on their knowledge and experience in terrestrial medicine. However, even after decades of spaceflight missions, we do not know if medications act the same in the spaceflight environment as they do on Earth. Aspects of the spaceflight environment (low gravity, radiation exposure, closed environment, stress) have been shown to alter human physiology. From first principles, some of these physiological changes could be expected to alter either pharmacokinetics (PK, how the body absorbs, distributes, metabolizes, and excretes administered medications) or pharmacodynamics (PD, receptors or signalling systems that are the targets of medication action). Anecdotal data has suggested that, at least for certain medications or indications, inflight medication efficacy is poor, and spaceflight crew use high doses, repeated doses, or even experience apparent treatment failures; fortunately none of these episodes has been serious enough to cause injury or evacuation from low-Earth orbit (LEO). However, in order to prepare for exploration missions where speedy evacuation to Earth may not be a possibility, the likelihood of unexpected medication action must be determined. New studies have yielded the first documented evidence of apparent treatment failures in low Earth orbit. These instances could be explained by alterations in PK or PD. The central hypothesis of this proposal is that the spaceflight environment alters human physiology in ways that affect how medications ultimately exert their actions, either by changing PK or PD. Specifically, we propose to: (1) test the hypothesis that PK following administration of a medication is different during spaceflight as compared to ground-based data from the same subject, and (2) test the hypothesis that medication action (PD) is different during spaceflight as compared to ground-based data from the same subject. We propose to test our hypothesis with two unrelated medications commonly used on spaceflight missions: a nonbenzodiazepine hypnotic, and an antibiotic. We propose to measure PK for 2 medications used by crewmembers in two different environments: on Earth and during a spaceflight mission. Circulating concentrations in the bloodstream will be measured over time after medication administration, permitting modeled estimation of absorption rates and metabolism/excretion rates. Same-subject comparisons using pre-flight and post-flight test sessions will permit measurement of same-subject variability that will provide a baseline for determination of clinically relevant changes during spaceflight. Spaceflight vs. ground differences in medication-associated effects on sleep, wakefulness, and several measures of cognitive function will be used as measures of pharmacodynamics. This study will determine if the LEO environment of the International Space Station (ISS) alters the actions of three common medications. A likely outcome, relevant for NASA missions, is optimized dosing recommendations for one or more of these medications. In more general terms, the results of this study will indicate the likelihood and magnitude of any spaceflight-associated PK or PD changes, and thus help to determine future research priorities. The general population, who also use these medications, are also likely to benefit from additional understanding of how these medications interact with the human body.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study will indicate the likelihood and magnitude of any spaceflight-associated PK or PD changes, and thus help to determine future research priorities. The general population, who also use these medications, are also likely benefit from additional understanding of how these medications interact with the human body.
	In May 2017, the PI was informed that the project was “deferred to a later date,” which effectively ended the project before data collection from crewmember subjects could begin. No new project activities were initiated after that date. We concluded the EEG evaluation that was in progress at that point. We are only able to report on preparatory optimization of analytical techniques and the EEG evaluation. No crew subject testing was ever begun. Part 1. Bioanalytical Method Development and Validation for the Simultaneous Quantification of Amoxicillin and Zaleplon in Human Plasma Specific aim: Determine spaceflight vs ground difference in PK profiles for two medications frequently used by crewmembers. In order to achieve this aim, a sensitive bioanalytical assay for simultaneous determination of amoxicillin and zaleplon concentrations should be validated. Amoxicillin and zaleplon in plasma were separated by using mass spectrometry. The bioanalytical method was validated according to the guidance provided by US Food and Drug Administration ((CDER) CfDEaR. Guidance for industry: bioanalytical method validation, U.S. Food and Drug Administration. 2013). The full validation for plasma including the assessment of specificity and selectivity, standard curves, accuracy and precision, extraction recovery, matrix effect, and sample stability. Selectivity and specificity were assessed by analyzing processed plasma samples from six different sources. All samples were spiked with amoxicillin, zaleplon, and internal standard, and the interference at corresponding retention time were compared against the LLOQ standards. Accuracy and precision (inter- and intra- day) for both analytes in plasma were evaluated. The extraction recovery at each QC levels were measured. Stability was evaluated for freeze-thaw, autosampler, short term (bench top), and long term (one month and three month). A rapid and sensitive LC-MS/MS method was developed and validated for simultaneous quantification of amoxicillin and zaleplon in human plasma. This is the first bioanalytical method to determine concentrations of these two analytes simultaneously. It provided a superior sensitivity with LLOQ as low as 0.2 ng/mL for zaleplon also remained good linearity at ULOQ as high as 15000 ng/mL for amoxicillin. The method can be further applied to in-flight and ground test sessions of pharmacokinetic study. Part 2: Effect of Zaleplon on Electroencephalogram Activity - A Pilot Study to Evaluate the Feasibility of Measuring Zaleplon Pharmacodynamics during Spaceflight Astronauts use medications during space missions to treat medical complaints. Most commonly, medications are taken to treat sleep problems, space motion sickness, pain, allergies, and sinus congestion [Wotring VE. Medication use by U.S. crewmembers on the International Space Station. FASEB J 2015;29:4417-23; Putcha L, Berens KL, Marshburn TH,

Task Progress:	Ortega HJ, Billica RD. Pharmaceutical use by U.S. astronauts on space shuttle missions. Aviat Space Environ Med 1999;70:705-8.]. Some of the medications that are administered during the missions are taken very frequently. For example, during space shuttle missions, sleep promoting drugs were taken on over 50% of the nights by the majority of crewmembers [Barger LK, Flynn-Evans EE, Kubey A, et al. Prevalence of sleep deficiency and use of hypnotic drugs in astronauts before, during, and after spaceflight: an observational study. Lancet Neurol 2014;13:904-12.]. Other medication classes that are used by more than 50% of astronauts include pain relievers for headache and congestion/allergy treatments [Wotring VE. Medication use by U.S. crewmembers on the International Space Station. FASEB J 2015;29:4417-23.]. The aim of this pilot study was to evaluate the feasibility of measuring the pharmacodynamic effect of zaleplon, one of the most frequently taken medications during spaceflight after oral administration of a standard dose of 5 mg and 10 mg of the drug in healthy subjects on Earth, with two different Electroencephalogram (EEG) devices. Ultimately, the goal of this study was to evaluate whether the pharmacodynamic effect of zaleplon can be detected via EEG during a space mission in the above mentioned doses. EEG is a measure of the mean electrical activity of different brain regions due to the sum of extracellular flows of a large group of neurons and has several advantages, making it a useful tool in drug research and development [Sakkalis V. Applied strategies towards EEG/MEG biomarker identification in clinical and cognitive research. Biomark Med 2011;5:93-105.]. The benefits of EEG are in line with many of the demands for ideal pharmacodynamic measures like being continuous, objective, sensitive, and reproducible [Mandema JW, Danhof M. Electroencephalogram effect measures and relationships between pharmacokinetics and pharmacodynamics of centrally acting drugs. Clin Pharmacokinet 1992;23:191-215.]. Furthermore, EEG recording is as a non-invasive and pain-free subject friendly method [Leiser SC, Dunlop J, Bowlby MR, Devilbiss DM. Aligning strategies for using EEG as a surrogate biomarker: a review of preclinical and clinical research. Biochemical pharmacology 2011;81:1408-21.]. EEG is mostly used in the evaluation of the pharmacodynamics of anesthetics, benzodiazepines, and opioids as well as in the diagnosis and assessment of neurologic disorders like epilepsy and sleep disorders [Bewernitz M, Derendorf H. Electroencephalogram-based pharmacodynamic measures: a review. Int J Clin Pharmacol Ther 2012;50:162-84.]. Several studies have shown that EEG can be used to measure the pharmacodynamics of drugs acting on the GABA-A/chloride-ion channel including zaleplon and zolpidem [Greenblatt DJ, Ehrenberg BL, Gunderman J, et al. Pharmacokinetic and electroencephalographic study of intravenous diazepam, midazolam, and placebo. Clin Pharmacol Ther 1989;45:356-65. ; Greenblatt DJ, Harmatz JS, von Moltke LL, et al. Comparative kinetics and dynamics of zaleplon, zolpidem, and placebo. Clin Pharmacol Ther 1998;64:553-61. ; Greenblatt DJ, Harmatz JS, von Moltke LL, et al. Comparative kinetics and response to the benzodiazepine agonists triazolam and zolpidem: evaluation of sex-dependent differences. J Pharmacol Exp Ther 2000;293:435-43. ; Greenblatt DJ, Legangneux E, Harmatz JS, et al. Dynamics and kinetics of a modified-release formulation of zolpidem: comparison with immediate-release standard zolpidem and placebo. J Clin Pharmacol 2006;46:1469-80.]. Hypothesis: EEG provides an objective and quantifiable measurement of the central effects of zaleplon. Specific aim 1: To describe the effect of zaleplon on brain activity over time by comparing changes in the ß EEG amplitude to baseline after oral administration of 5 mg and 10 mg zaleplon. Specific aim 2: To compare two different EEG devices in regards to ß frequency band changes after zaleplon administration, data quality, comfort, and user friendliness. Specific aim 3: To evaluate the feasibility of determining the pharmacodynamic effect of zaleplon via EEG during a space mission after administration of 5 mg and 10 mg zaleplon orally. This pilot study has several limitations that need to be taken into account before conclusions can be drawn. First and foremost, the sample size (N=3) is very small. Since the between-subject variability during the study was rather high, it is difficult to conclusively interpret the results. In addition, the study was not placebo controlled making it even more difficult to evaluate if changes observed were caused by zaleplon or other factors. However, other EEG studies have shown that the placebo response is small and that if changes in relative ß EEG amplitude after placebo administration are observed, they are usually opposite to the changes observed after administration of benzodiazepine agonists.
Bibliography Type:	Description: (Last Updated: 12/24/2019)
Articles in Peer-reviewed Journals	Kast J, Yu Y, Seubert CN, Wotring VE, Derendorf H. "Drugs in space: Pharmacokinetics and pharmacodynamics in astronauts." Eur J Pharm Sci. 2017 Nov 15;109S:S2-S8. Epub 2017 May 19. Review. https://doi.org/10.1016/j.ejps.2017.05.025 ; PubMed PMID: 28533143 , Nov-2017